

SimpleScalar Installation Guide

The SimpleScalar tool set is a system software infrastructure used to build modeling applications for program performance analysis, detailed microarchitectural modeling, and hardware-software co-verification. Using the SimpleScalar tools, users can build modeling applications that simulate real programs running on a range of modern processors and systems.

This document will guide you through the installation procedure for the SimpleScalar toolset on a linux machine. The contents here are from [1-4] with a bit of modification.

1. Download the all-in-one archive file SimpleScsalar.tar.gz from the class webpage or BBL.
2. Un-tar the file

```
$ tar -xzf SimpleScalar.tar.gz
```

This command will create a directory 'SimpleScalar' which contains five archive files.

3. Set some environment variables. Below is an example assuming your current directory is /home/john (type 'pwd' to check the current directory). Also we assume the host machine uses the little endian representation (if not, refer to [4]).

```
$ export IDIR=/home/john/SimpleScalar
```

```
$ export HOST=i686-pc-linux
```

```
$ export TARGET=sslittle-na-sstrix
```

4. Check if your machine has the following packages installed.

- 1) build-essential

- 2) flex

- 3) bison

5. Installing SimpleTools

Just un-tar the simpletools package, and remove the gcc directory.

```
$ cd $IDIR
```

```
$ tar -xvfz simpletools-2v0.tgz
```

```
$ rm -rf gcc-2.6.3
```

6. Installing SimpleUtils

Un-tar the package and fix an error.

- a. Un-tar the package

```
$ cd $IDIR
```

```
$ tar -xvfz simpleutils-990811.tar.gz
```

```
$ cd simpleutils-990811
```

- b. Open `ldlex.l` file in `ld` directory, and replace `yy_current_buffer` with `YY_CURRENT_BUFFER`
 - c. Install

```
$ ./configure --host=$HOST --target=$TARGET --with-gnu-as --with-gnu-ld --
prefix=$IDIR
$ make
$ make install
```
7. Installing the simulator
 - a. Un-tar the package

```
$ cd $IDIR
$ tar -xvfz simplesim-3v0d.tgz
```
 - b. Open `simplesim-3.0\Makefile` file and replace `“-O0”` with `“-O”`
 - c. Install

```
$ make config-pisa
make
```
 - d. Test if the installation is successful

```
$ cd $IDIR/simplesim-3.0
$ ./sim-safe tests/bin.little/test-math
```
8. Installing gcc cross-compiler
 - a. Un-tar the package

```
$ cd $IDIR
$ tar -xvfz gcc-2.7.2.3.ss.tar.gz
$ cd gcc-2.7.2.3
$ ./configure --host=$HOST --target=$TARGET --with-gnu-as --with-gnu-ld --
prefix=$IDIR
```
 - b. Modify some files
 - i. Open `Makefile` file, go to line 130, and append `-I/usr/include` to the end of the line
 - ii. Open `protoize.c`, go to line 60, and replace `#include <varargs.h>` with `#include <stdarg.h>`
 - iii. Open `obstack.h`, go to line 341, and replace `next_free)++` with `next_free++`
 - c. Type the following three commands
 - i. `cp ./patched/sys/cdefs.h ../sslittle-na-sstrix/include/sys/cdefs.h`
 - ii. `cp ../sslittle-na-sstrix/lib/libc.a ../lib/`
 - iii. `cp ../sslittle-na-sstrix/crt0.o ../lib/`

d. Build the compiler

Type the following command, and it will stop with some error messages

```
$ make LANGUAGES="c c++" CFLAGS="-O" CC="gcc"
```

Open insn-output.c file, append \ to the end of lines 675, 750 and 823.

Type "make LANGUAGES..." again, and it will stop again with different error messages

Open cxxmain.c file and remove the two lines 2978 and 2979

Finally, type the following

```
$ make LANGUAGES="c c++" CFLAGS="-O" CC="gcc"
```

```
$ make LANGUAGES="c c++" CFLAGS="-O" CC="gcc" install
```

e. Done

Now everything is ready. Before we finish, let us set some environment variables

```
$ export PATH=$PATH:$IDIR/bin:$IDIR/simplesim-3.0
```

References

[1] <http://simplescalar.com/>

[2] http://www.cse.iitd.ernet.in/~cs5070217/csl718/assignment1/ss_install_instructions.html

[3] http://www.kth.se/polopoly_fs/1.225471!/Menu/general/column-content/attachment/simplescalar-installation-instructions.pdf

[4] http://simplescalar.com/docs/users_guide_v2.pdf